

Detailed Daily Task Report — June 13, 2026

Project: StudentKhabri CRM & Chatbot

Developer: Pratik Choudhuri

1. Infrastructure & Subdomain Hosting configuration

- **Subdomain Creation:** Initiated the deployment process by establishing a dedicated subdomain specifically for the chatbot interface (`chatbot.studentkhabri.com`).
- **NGINX Reverse Proxy Setup:** Configured an NGINX server block to capture inbound traffic to the new subdomain and securely proxy it to the internal application process (managed via PM2).
- **SSL Certificate Generation:** Verified existing SSL coverage and executed Let's Encrypt (`certbot`) to provision and bind valid SSL/TLS certificates for the subdomain, ensuring end-to-end encrypted HTTPS connections.

2. Frontend Debugging and Optimization

- **Static Asset Resolution:** Investigated console errors where the server was responding with HTTP 404 (Not Found) for critical resources.
- **Path Correction:** Fixed the build and routing paths to ensure that the frontend successfully loads all required styles, scripts, and assets when served through the new NGINX proxy setup.

3. Chatbot Logic & Handoff Engineering

- **Human-Handoff Flow:** Overhauled the conversational logic that triggers when a lead is successfully assigned to a human representative in the CRM.
- **User Experience (UX):** Instead of terminating the chat, the bot now dispatches a reassuring prompt: *"We have assigned you to a representative. You can still get more info from this chat or wait till our representative calls you."* This keeps the user engaged and allows them to ask standard FAQ questions while waiting.

4. API Rate Limiting & Error Handling

- **Backend Constraints:** Configured the application to monitor and respect the upstream LLM/Provider API rate limits.
- **Frontend Visibility:** Engineered the frontend chat interface to gracefully catch HTTP 429 (Rate Limit Exceeded) errors or limit hits, translating them into a clear, user-facing warning directly within the chat UI. This prevents silent failures and drastically improves the user experience during high-traffic periods.

Next Steps

- Continue to monitor the NGINX logs for the new `chatbot` subdomain to ensure traffic is flowing smoothly.
- Test the rate-limit UI under simulated heavy load to ensure warnings trigger correctly.